

		Intensity $\propto$ (amplitude) ² , if intensity is one ninth, amplitude is one third
22	B	Slit separation, $d = \frac{1}{500} = 0.002 \text{ mm} = 2.00 \times 10^{-6} \text{ m}$ $d \sin \theta = n \lambda$ $n = \frac{d \sin \theta}{\lambda} = \frac{2.00 \times 10^{-6} \sin(90^\circ)}{600 \times 10^{-9}} = 3.33$ Maximum observable is 3rd order. Therefore, answer is 7.
23	A	For one of the bright light, the first order minima is located at an angle θ where